

EDUCATION

National University of Singapore (NUS)

Aug 2019 – Dec 2023

- Bachelor of Computing (Computer Science), Honours
- Second Major in Statistics & Minor in Mathematics
- CAP: 4.68 / 5.0 (1st Upper Class Honours with Highest Distinction)

RESEARCH EXPERIENCE

Interactive Visual Grounding of Ambiguous Natural Language Instructions

Jan 2023 – Dec 2023

Undergraduate Final Year Project, National University of Singapore

- Developed an interactive, attribute-guided disambiguation system for robotic placement tasks, enabling robots to resolve ambiguities in natural language instructions.
- Built an entropy-based query selection mechanism, optimizing the robot's questioning strategy to extract the most informative responses.
- Achieved a significant improvement in grounding accuracy, with experiments demonstrating enhanced success rates in disambiguating object placement tasks.

NUS Bayesian Optimization Research Group, Research Assistant

Aug 2022 – Dec 2022

- Performed metamodeling for computer simulations and algorithmic development for simulation optimization to solve real-world problems
- Processed high dimensional real-world data using machine learning techniques

NUS-NCS Joint Laboratory for Cyber Security, Research Assistant

Aug 2022 – Nov 2022

- Designed and implemented efficient and robust systems and machine learning algorithms aimed at data security and privacy solutions

INDUSTRY AI EXPERIENCE

UBS, Applied AI Engineer

Feb 2024 – Present

LLM-Enhanced Client Behavior Modeling and Business Insights

Jan 2025 – Present

- Developed a hybrid statistical-LLM pipeline to model client financial behavior from multi-source signals including account flows, holdings, and trading activity.
- Constructed latent behavioral representations to characterize saving and trading patterns, enabling interpretable client profiling for downstream advisory decision support.

Multilingual Speech-to-Text and Audio Intelligence Systems

Jun 2025 – Feb 2026

- Built an end-to-end multilingual speech-to-text system supporting English, Mandarin, and Cantonese, integrating language identification, speaker diarization, transcription, and translation.
- Investigated LLM-based synthetic data generation and domain adaptation techniques to fine-tune Whisper models for finance-specific terminology and accented speech robustness.

LLM Web Agent for Automated Ticket Resolution

Jun 2024 – Dec 2024

- Formulated an LLM-based web agent framework for cross-platform ticket automation, leveraging few-shot learning from successful trajectories and failure-case feedback.
- Studied feedback-driven refinement mechanisms to improve agent robustness in noisy environments, reducing manual support workload by ~50%

Orchestrable AI Framework & Generative AI Knowledge Systems

Feb 2024 – Jun 2024

- Developed key components of an orchestrable AI framework leveraging large language models (LLMs) to improve internal workflow efficiency and task automation.
- Designed and implemented a GenAI-powered graph knowledge system to extract implicit market signals from public financial data, enabling improved client insight and decision making.

Tencent, AI Engineer

Jun 2021 – Jun 2022

ART (Availability, Robustness, Trustworthiness) Model Evaluation Framework

Jun 2021 – Jun 2022

- Developed evaluation methodologies for assessing large-scale AI models in terms of availability, robustness, and trustworthiness in production systems serving 10M+ users.
- Designed black-box adversarial testing strategies using data augmentation to systematically probe model weaknesses and improve robustness under distribution shifts.

PATENTS & CERTIFICATIONS

A Method for Applying Data Augmentation in the Field of Essay Correction

Jun 2022

Patent Granted

- Developed an automated data augmentation method to enhance AI-driven essay grading models by simulating real-world error patterns.
- Developed a multi-level adversarial attack framework to generate diverse training samples, enhancing model robustness and accuracy.
- Patent Link: <http://epub.cnipa.gov.cn/patent/CN114626365A>

Tencent Cloud Practitioner Certificate

Dec 2023

- Certificate Link: https://drive.google.com/file/d/1Tp2mACa_7q3el1E8dZj1CLE1DRr6fKUI

TEACHING ASSISTANT EXPERIENCE

Teaching Assistant – Programming Methodology II

Aug 2020 – May 2021

Department of Computer Science, National University of Singapore

- Supervised lab sessions, provided guidance on Object-Oriented and Functional Programming concepts, and ensured students grasped key programming paradigms.
- Prepared teaching materials, promptly addressed student queries, and offered academic support to enhance learning outcomes.

SCHOOL PROJECTS

Face Mask Detection for Disease X: A Probabilistic Approach

Jan 2023 – Jun 2023

CS5340: Uncertainty Modelling, National University of Singapore

- Developed a Bayesian deep learning model with denoising variational autoencoders (DVAE) to improve robustness in noisy surveillance images.
- Implemented Bayesian convolutional neural networks (BCNNs) to quantify uncertainty, improving decision reliability in mask detection.

Reinforcement Learning from Human Feedback in ChatGPT for Conversational Abilities

CS4246: AI Planning and Decision Making, National University of Singapore

Jan 2023 – Jun 2023

- Investigated Reinforcement Learning from Human Feedback (RLHF) and its impact on aligning ChatGPT's responses with human preferences.
- Analyzed Proximal Policy Optimization (PPO) in fine-tuning ChatGPT, evaluating its effectiveness in improving response quality.

Aspect-Based Sentiment Analysis with Enhanced Graph Convolutional Networks

CS4248: Natural Language Processing, National University of Singapore

Aug 2022 – Dec 2022

- Developed an enhanced graph convolutional network (GCN) with optimized dependency tree structures for improved sentiment polarity classification.
- Incorporated affective information from SenticNet into the adjacency matrix, refining sentiment analysis accuracy.

Orbital – Software Engineering Project (Advanced Certificate)

Jun 2020 – Aug 2020

- Certificate Link: <https://credentials.nus.edu.sg/f5e9b230-732e-46fc-afd9-e1bed913ceef>

CO-CURRICULAR ACTIVITIES

Core Team Member, We Are Hear (Singapore)

Aug 2022 – Present

- Support mental health awareness initiatives by coordinating listening sessions and managing social media for a non-profit promoting empathetic, stigma-free community support.

Technical Volunteer, Wicare (Singapore)

Aug 2022 – Dec 2022

- Produced a review video for the annual event, reducing production costs by ~\$2,000 and supporting outreach and engagement for volunteers and participants.

TECHNICAL SKILLS

- AI & ML: LLM, generative AI, statistical machine learning, model evaluation and robustness.
- LLM Systems & Agents: LangChain, CrewAI, DSPy, Agency-Swarm, Azure OpenAI; experience in building agentic and RAG-based applications.
- Programming: Python, Java, C++; strong software engineering and system design background.
- Data Science & Statistics: Probability, statistical inference, and applied data modelling.
- ML Frameworks: PyTorch, NumPy, OpenCV; experience in training, finetuning, and deploying production-grade deep learning models.
- Algorithms: Data structures and algorithms with practical application in ML and large-scale systems.